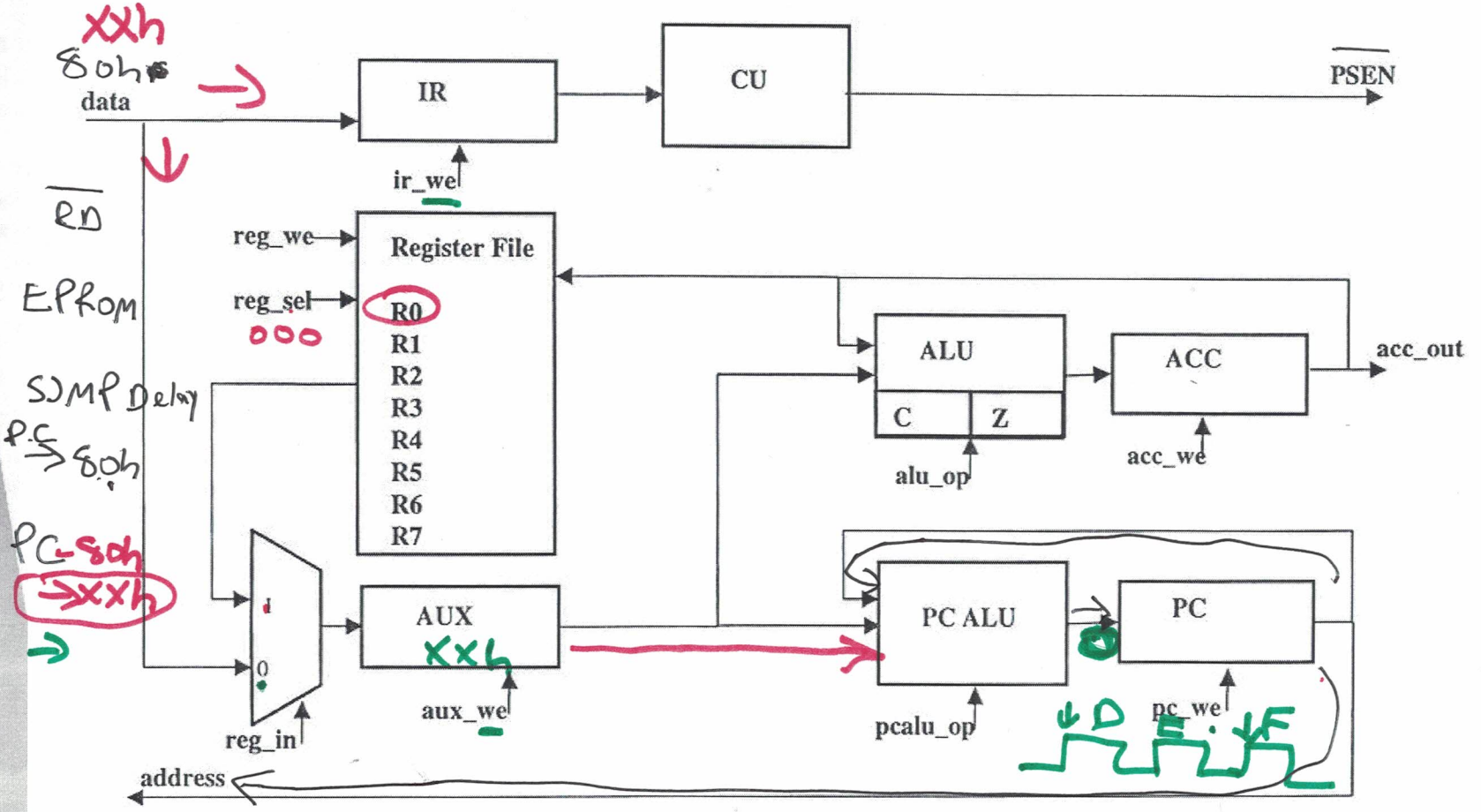
WIMP51 Architecture



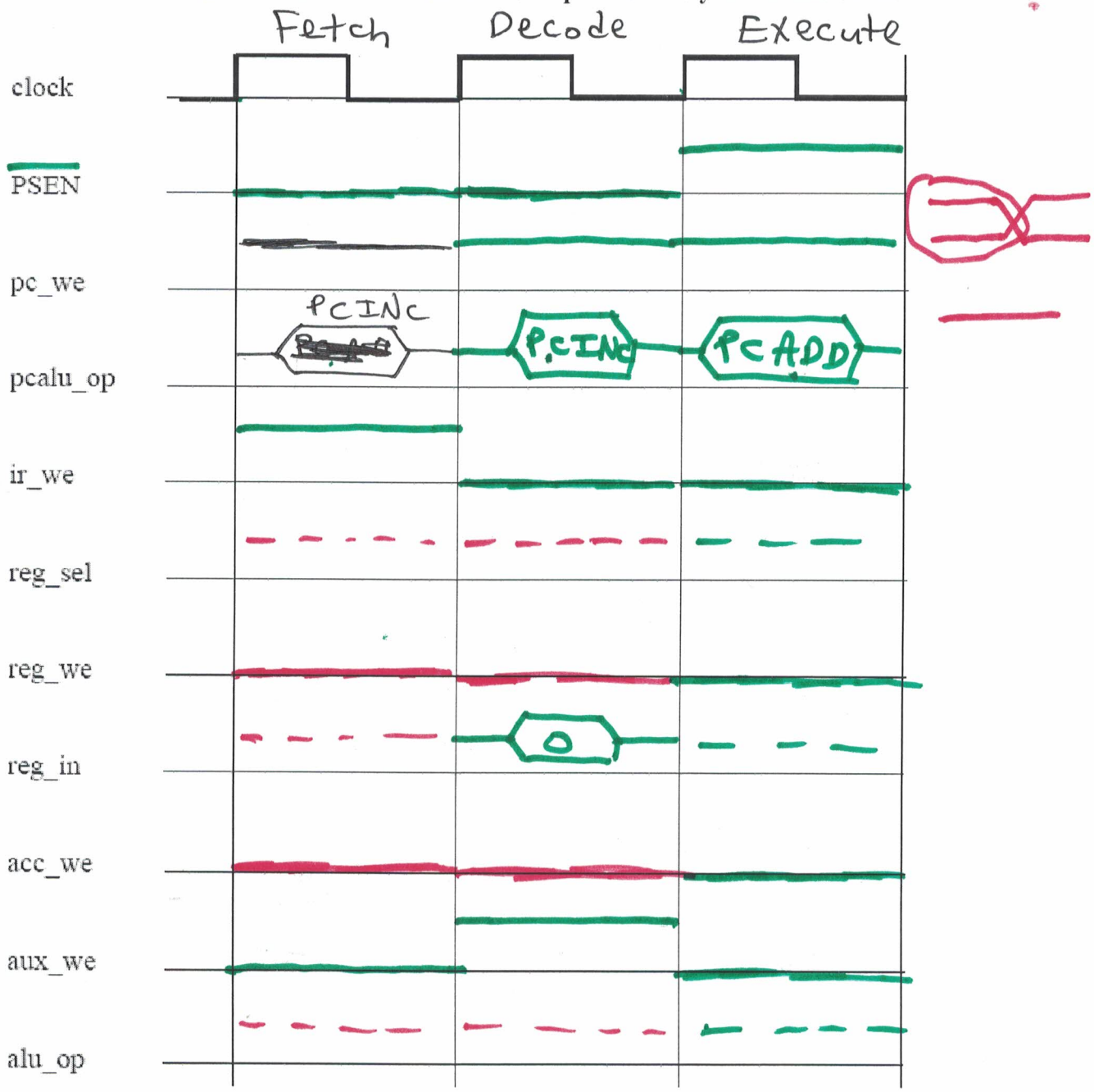
Wimp51 Instructions

MOV A, #D	01110100	dddddddd	A<=D
ADDC A, #D	00110100	dddddddd	C,A<=A+D+C
MOV Rn, A	11111nnn		Rn<=A
MOV A, Rn	11101nnn		A<=Rn
ADDC A, Rn	00111nnn		C,A<=A+Rn+C
ORL A, Rn	01001nnn		A<=A or Rn
ANL A, Rn	01011nnn		A<=A and Rn
XRL A, Rn	01101nnn		A<=A xor Rn
SWAP A	11000100		$A<=A_{(3-0)}\&A_{(7-4}$
CLR C	11000011		C<=0
SETB C	11010011		C<=1
SJMP rel	10000000	aaaaaaaa	PC<=PC+rel+2
JZ rel	01100000	aaaaaaaa	PC<=PC+rel+2 if Z

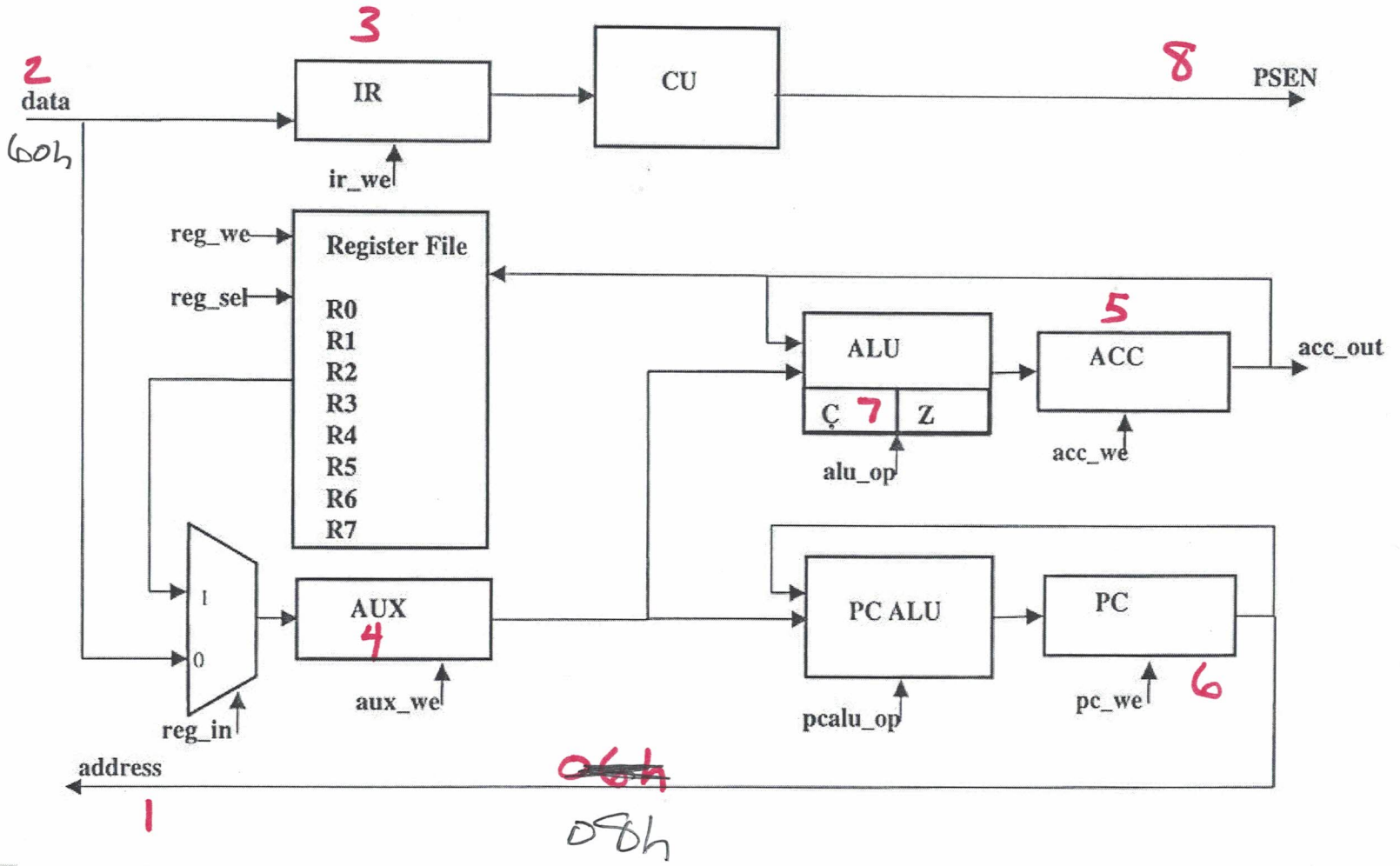
Draw the timing diagram for the following control signals on the diagram below when the WIMP51 is executing the instruction SJMP Delay Label the fetch, decode and execute parts of the instruction cycle on the clock portion of the signal below.

The dotted lines have been drawn as a graphical reference only and do not reflect the value of the corresponding signal. If you think a particular control signal is of no significance during a particular time period, indicate this opinion by using a **dashed** line to denote the signal as a "don't care" for that time period. Give "command" values for pcalu_op and alu_op; for example, indicate the value of "pcalu_op" as "PC_INC" or "PC_ADD" rather than representing this command by a numeric code. If you are not sure about a particular control signal, make an educated guess, indicate that the value you have marked is a guess, and explain why you made this particular guess. Remember, PSEN/ pulses low whenever a byte is being fetched from code memory.

The WIMP51 architecture schematic is available in the notes provided to you for this exam.



WIMP51 Architecture

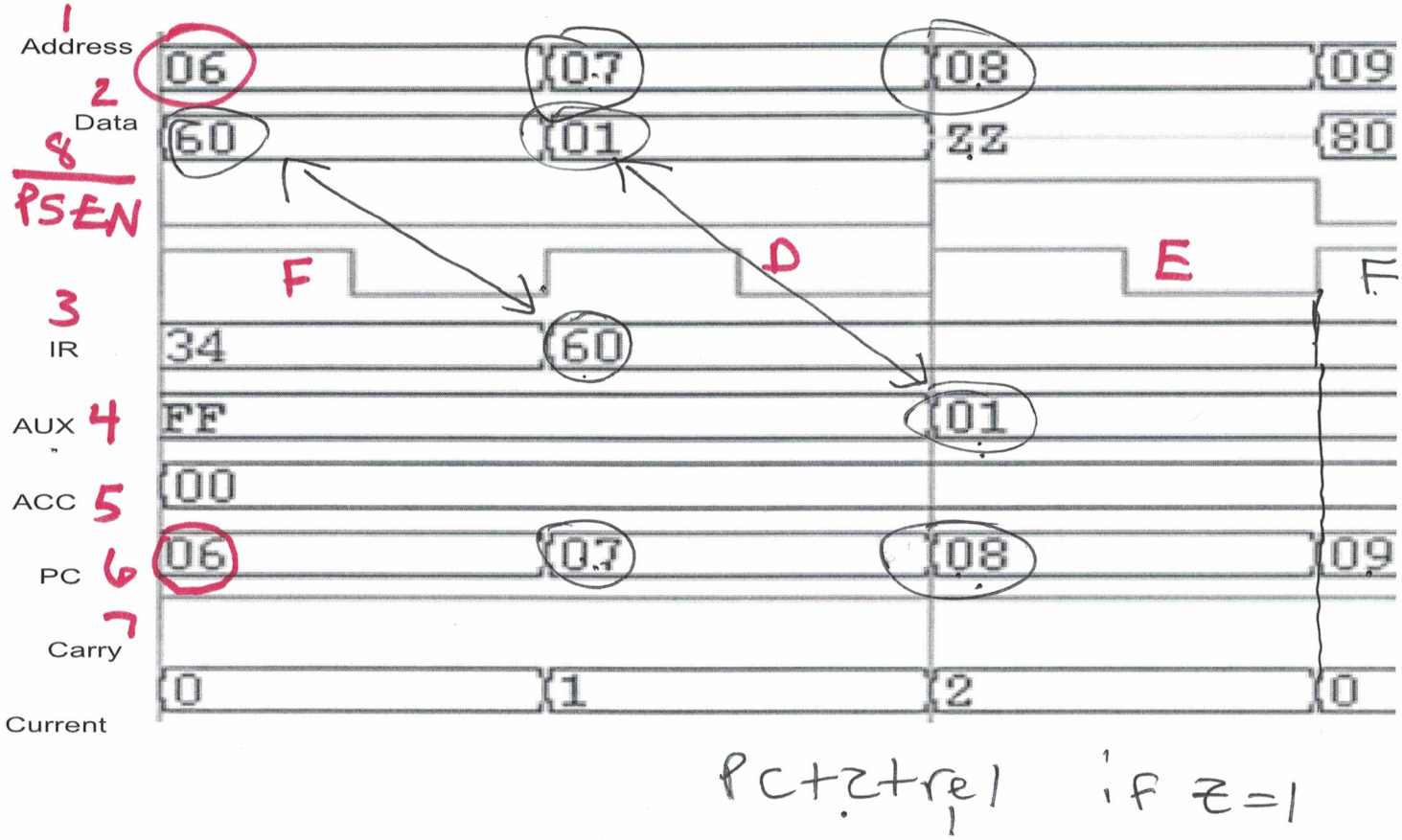


Wimp51 Instructions

MOV A, #D	01110100	dddddddd	A<=D
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ADDC A, Rn	00111nnn		C,A<=A+Rn+C
ORL A, Rn	01001nnn		A<=A or Rn
ANL A, Rn	01011nnn		A<=A and Rn
XRL A, Rn	01101nnn		A<=A xor Rn
SWAP A	11000100		$A<=A_{(3-0)}\&A_{(7-4}$
CLR C	11000011		C<=0
SETB C	11010011		C<=1
SJMP rel	10000000	aaaaaaaa	PC<=PC+rel+2
JZ rel	01100000	aaaaaaaa	PC<=PC+rel+2 if Z

60h

Given the following timing signals for the WIMP51 for an instruction cycle. For the Current State signal, note that 0 = fetch state, 1 = decode state, 2 = execute state.



Determine the instruction being executed during the instruction cycle.

JZ rel

JZ Funct
60h rel

What is the PC after executing this instruction?

PC=0X09